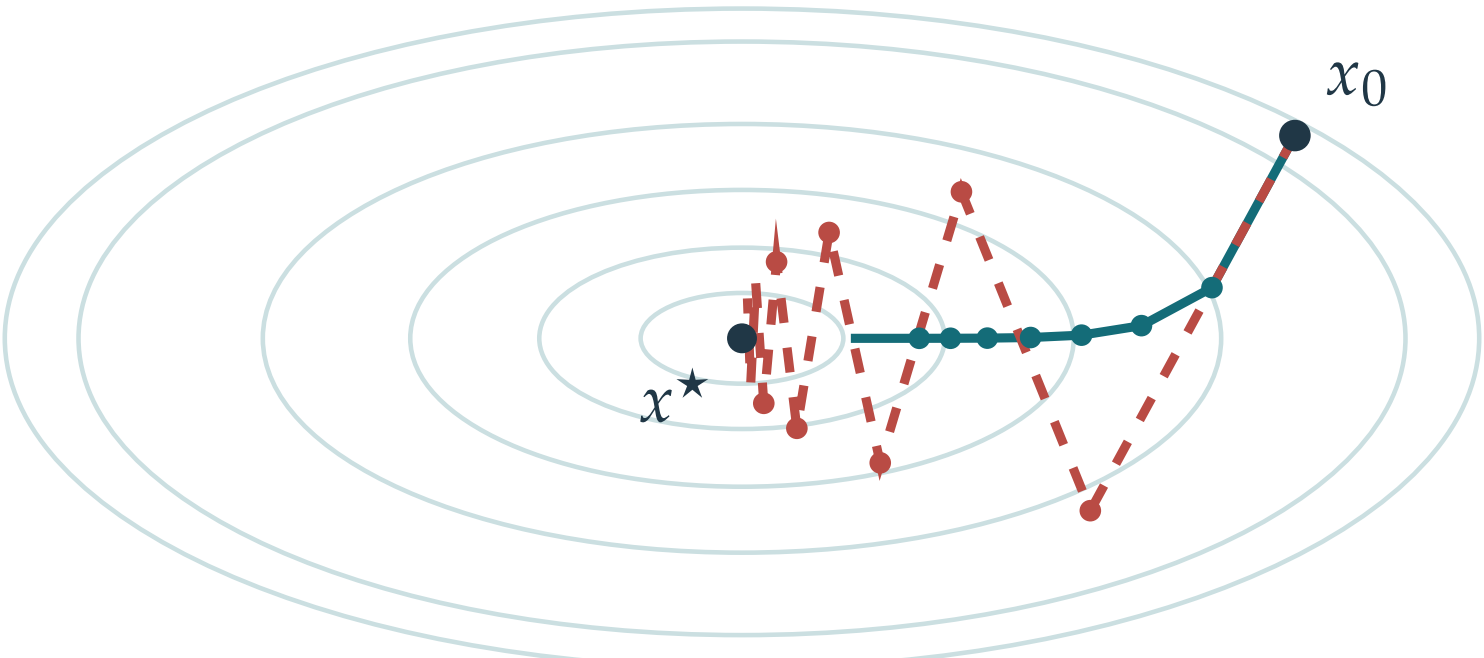
$$f(x) = \tfrac{1}{2}(x_1^2 + 5x_2^2)$$



$\tau = 0.15$ $\tau = 0.37$

A larger step can oscillate
across the narrow direction